EXECUTIVE DATA ANALYTICS REPORT: SUPERSTORE PERFORMANCE

Quantitative Analysis on Revenue Drivers, Customer Segments, and Regional Profitability

- **Prepared By:** Vipanjeet Kaur

- **Date:** August 25, 2026

Executive Summary

Volume vs. Profit Disparity:

- Technology acts as the primary bottom-line driver, producing $39,774.07 in total profit ($86.65 per transaction).
- Furniture suffers severe margin erosion, returning only $12.38 in profit per order due to excessive discounting (17.69%).

Customer Segment Ranking:

- Corporate accounts represent the most lucrative tier, yielding the highest average transaction value ($261.17) and highest profit per order ($39.09).
- Consumer generates high order volume but lower average margin efficiency ($26.94 per transaction).

Regional Parity & Policy:

- Two-sample t-testing confirms no statistically significant difference in transactional profitability between East ($26.29) and West ($29.88) regions ($p = 0.7657$).
- The Central region lags in net performance due to aggressive average discount rates (22.21%).

Dataset Overview & Descriptive Metrics

Financial Metric	Mean	Median	Std Dev	Total Sum
Sales Revenue	$235.49	$54.42	$500.57	$609,205.86
Profit Margin	$31.62	$8.67	$211.96	$81,795.23
Discount Rate	15.65%	0.00%	20.64%	N/A

Key Insights:

- **Transaction Scope**: Dataset contains 2,587 transactions across 21 business variables.
- **Positive Revenue Skew:** Mean sales ($235.49) significantly exceed median sales ($54.42), proving revenue is driven by a small volume of high-value purchases.
- **Profit Volatility:** Extreme profit dispersion ranging from -$6,599.98 to +$8,399.98 highlights high exposure to discounting risks.

Customer Segment Profiling

Customer Segment	Total Revenue	Avg Order Value	Avg Profit / Order
Corporate	$207,106.41	**$261.17**	**$39.09**
Home Office	$105,235.48	$225.83	$32.25
Consumer	**$296,863.97**	$223.54	$26.94

Key Insights:

- **Consumer Segment:** Serves as the high-volume engine, bringing in the highest total revenue ($296.86k) but generating the lowest profit per order ($26.94).
- **Corporate Segment**: Demonstrates the highest operational efficiency, leading with an average order value of $261.17 and the highest profit contribution per transaction ($39.09).
- **Home Office Segment**: Sits in the middle with $105.24k in total sales and an average order profit of $32.25.

Regional & Category Economics

Product Categories

Category	Total Sales	Total Profit	Avg Profit / Order	Avg Discount
Technology	$215,400.00	**$39,774.07**	**$86.65**	13.20%
Office Supplies	$198,120.00	$35,061.23	$22.39	15.80%
Furniture	$195,685.86	**$6,959.93**	**$12.38**	**17.69%**

Category Key Insights:

- **Technology is the main profit engine:** Delivers $86.65 profit per order, nearly 7x higher than Furniture.
- **Furniture suffers severe margin erosion**: High discounting (17.69%) drops per-order profit to just $12.38 despite generating nearly $195.7k in sales.

Regional & Category Economics

Regional Breakdown

Region	Total Sales	Total Profit	Avg Discount
West	$187,412.10	**$24,051.55**	**11.72%**
East	$156,210.80	$20,141.57	14.28%
Central	$141,105.40	$19,899.19	**22.21%**
South	$124,477.56	$17,702.92	15.17%

Regional Key Insights:

- **West leads in margin discipline**: Maintains the lowest average discount (11.72%), yielding the highest overall profit.
- **Central region lags in yield**: Heavy promotional discounting (22.21%) significantly depresses net profit margin.

Inferential Statistics & Hypothesis Testing

95% Confidence Interval (Sales)

Parameter	Value
Sample Size (n)	200 Orders
Sample Mean	$219.27
Standard Error	$32.59
95% Confidence Interval	**[$155.41, $283.14]**

Key Insights:

- **Population Validation**: We are 95% confident that the true population mean order value lies between $155.41 and $283.14.
- The actual overall mean sales ($235.49) falls cleanly inside this range, confirming strong sample accuracy.

Inferential Statistics & Hypothesis Testing

Two-Sample t-Test (East vs. West Profit)

t-Test Parameter	Value
Tested Difference	East ($26.29) vs. West ($29.88)
t-Statistic	**-0.2981**
p-Value	**0.7657** (\alpha = 0.05)
Decision	**Fail to Reject the null hypothesis**

Key Insights:

- **No Significant Difference**: Since the p-value (0.7657) is much larger than 0.05, there is no statistically significant difference in unit profitability between East and West.
- **Strategic Takeaway**: Regional performance gaps stem from volume, not per-transaction profitability.

Strategic Recommendations & Action Plan

Priority	Strategic Initiative	Key Action Item	Expected Impact
High	**Furniture Discount Cap**	Enforce a strict **10% maximum discount ceiling** on all Furniture products.	Prevents severe margin erosion; recovers profit margins above $35.00/order.
High	**B2B Corporate Focus**	Shift 30% of sales and marketing resources toward acquiring **Corporate accounts**.	Capitalizes on highest AOV ($261.17) and highest profit contribution ($39.09/order).
Medium	**Central Region Pricing Alignment**	Restructure Central region promotional policies to match the West region baseline (11.72%).	Reclaims lost revenue in the Central region without sacrificing transaction volume

Strategic Takeaway

- **Core Business Impact**: By capping discounting in low-margin categories like Furniture and focusing sales efforts on high-yield B2B Corporate clients, the business can significantly increase net profitability without needing to increase total transaction volume.

THANKYOU

Any questions or insights?